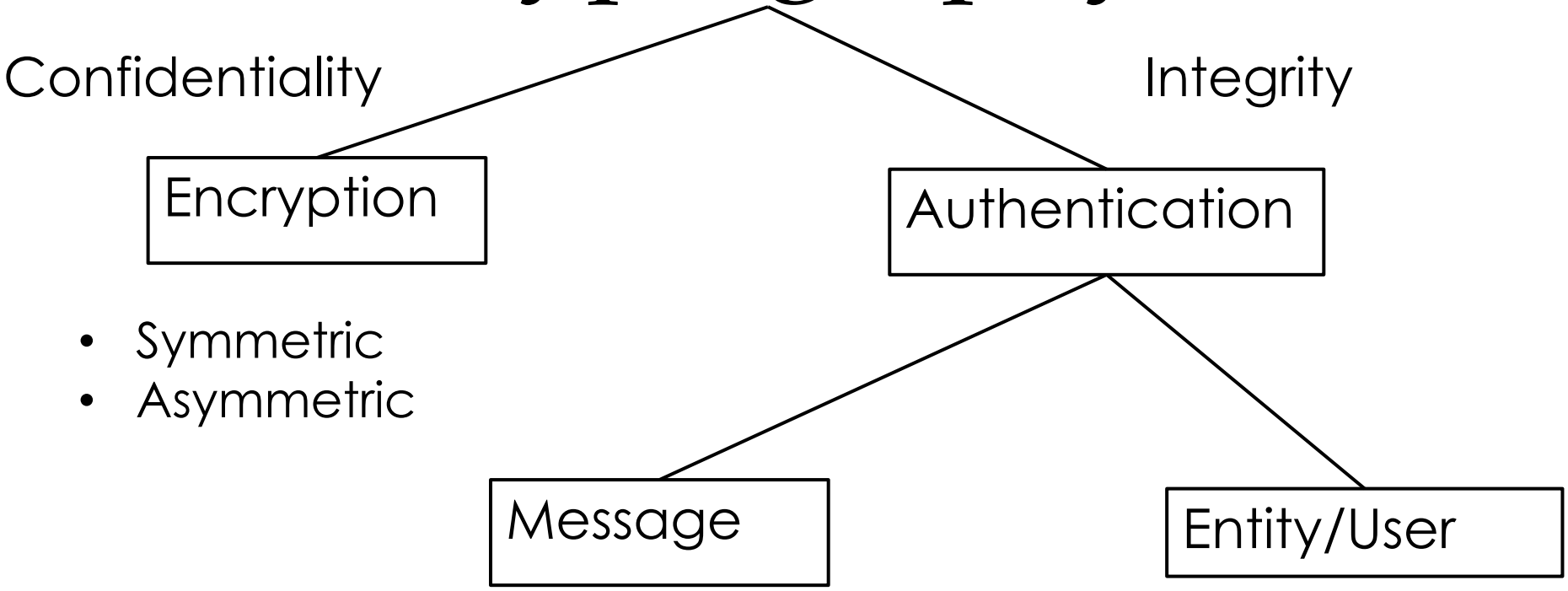
بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ

اَلْحَمْدُ لِلّٰهِ رَبِّ الْعَالَمِيْنَ

وَالصَّلٰوةُ وَالسَّلَامُ عَلٰى خَاتَمِ النَّبِيِّيْنَ

Entity/User Authentication

Cryptography



- Symmetric
- Asymmetric

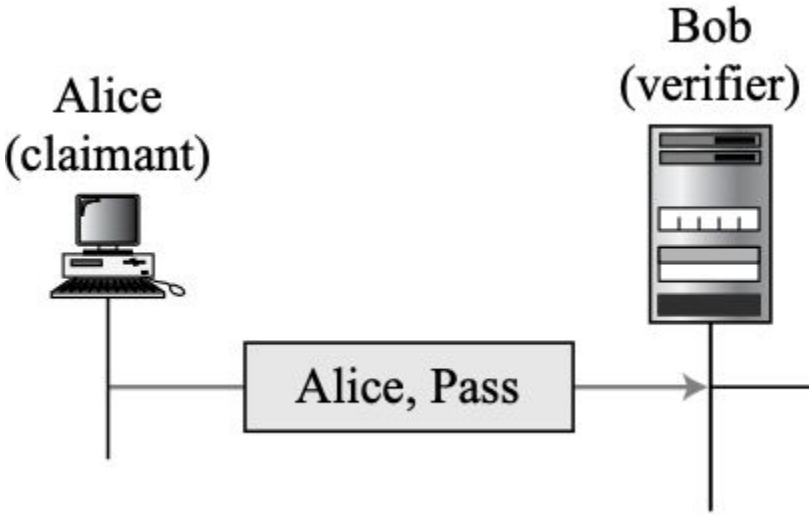
- Authentication Codes
- Digital SIgnatures

- Challenge Response
- Zero Knowledge

User Authentication

Entity / User / Endpoint Authentication

To prove **claimants** identity to a **verifier**



Different from message integrity as:

1. Does not have a fixed(usually public) content to authenticate
2. Has to be in real time with both parties online simultaneously
3. Once for entire subsequent session

Session identified using a secret session id e.g. as cookies encrypted within https blocks to avoid hijacking

Mechanism

User authentication is based on a secret that the claimant has:

It can be either

Knowledge: Passwords, PINs

Possession: Smart cards, Token devices

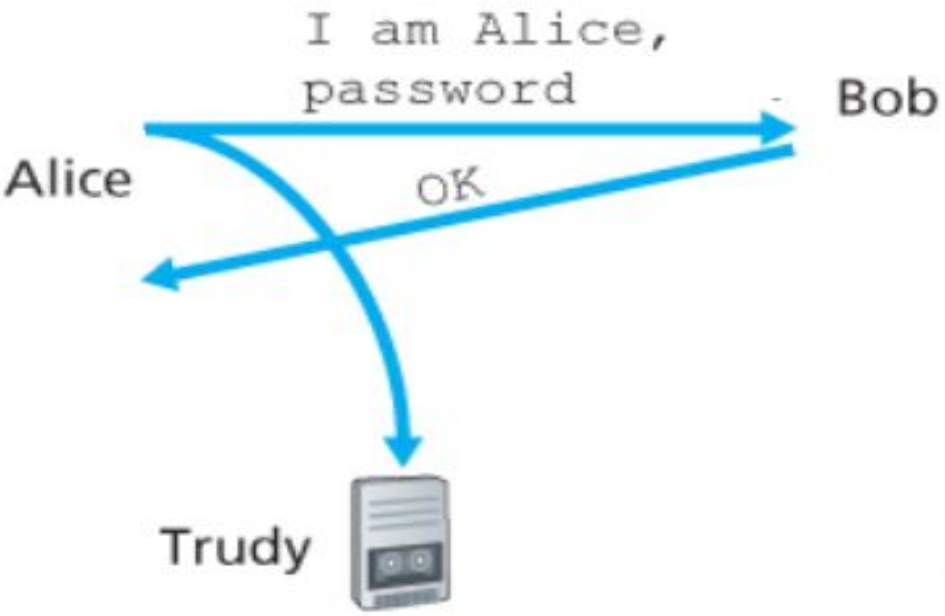
Inherent: Biometric, e.g.retina, fingerprint

Replay Attack

Authentication uses a secret known to claimant

But any attacker sniffing on insecure channel can record and replay the secret if shared

Authentication must avoid replay that is effective even for encrypted/signed messages





Nonce

Upon authentication request, a new, randomly generated number is sent to claimant as challenge from verifier

to be encrypted/decrypted/signed by claimant

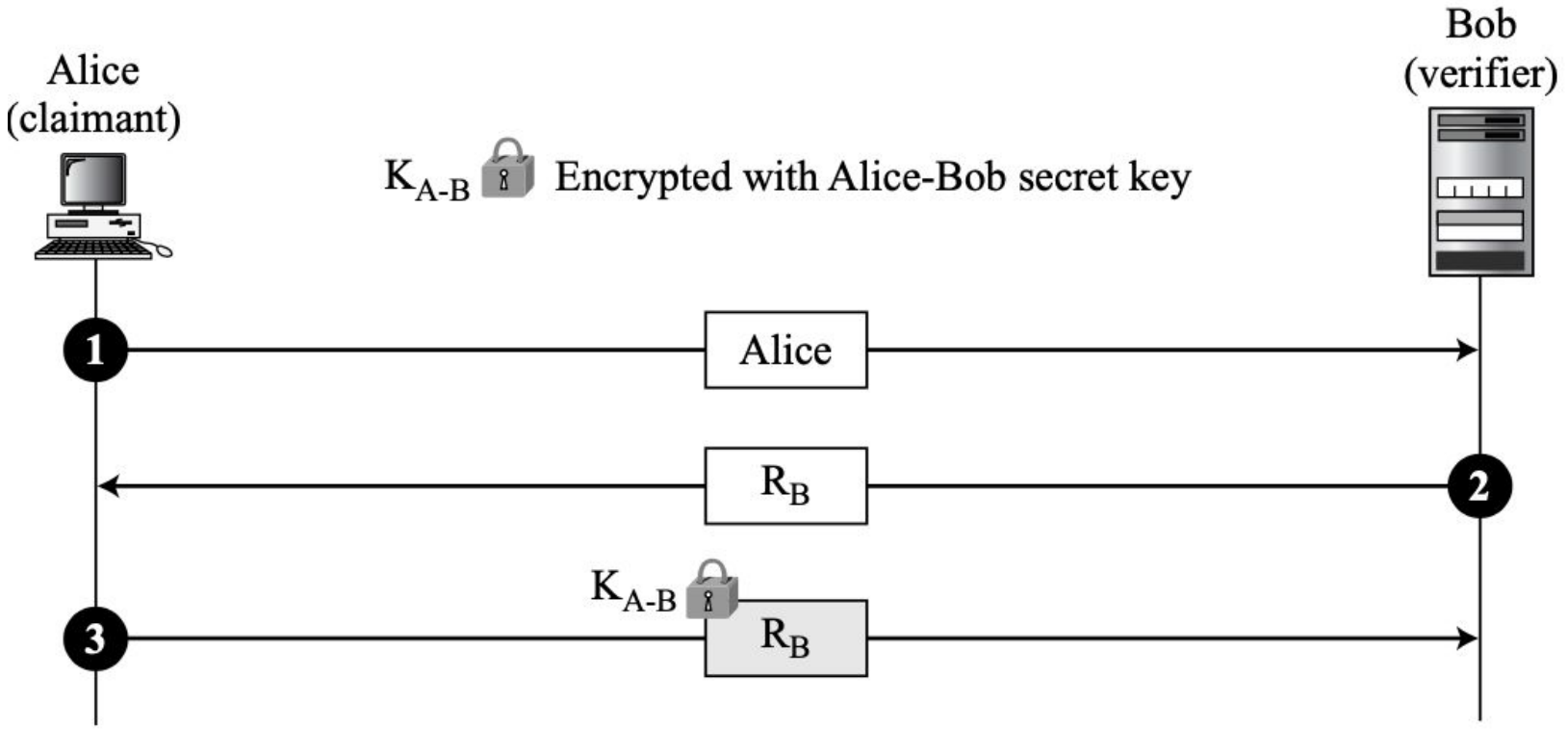
Using a secret (in a symmetric or asymmetric manner)

Nonce is new each time hence not reusable

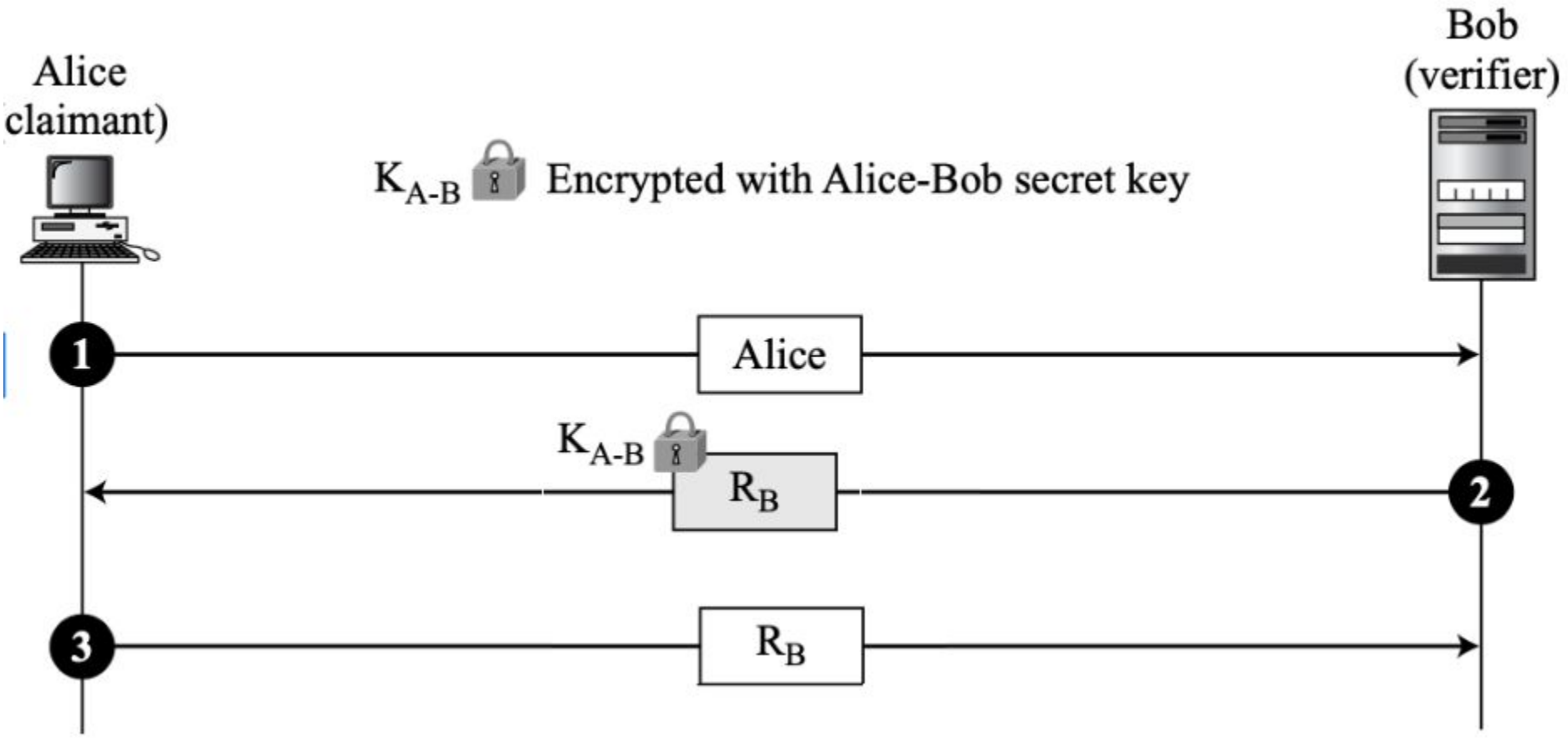
Symmetric Encryption

RB being random nonce sent by verifier

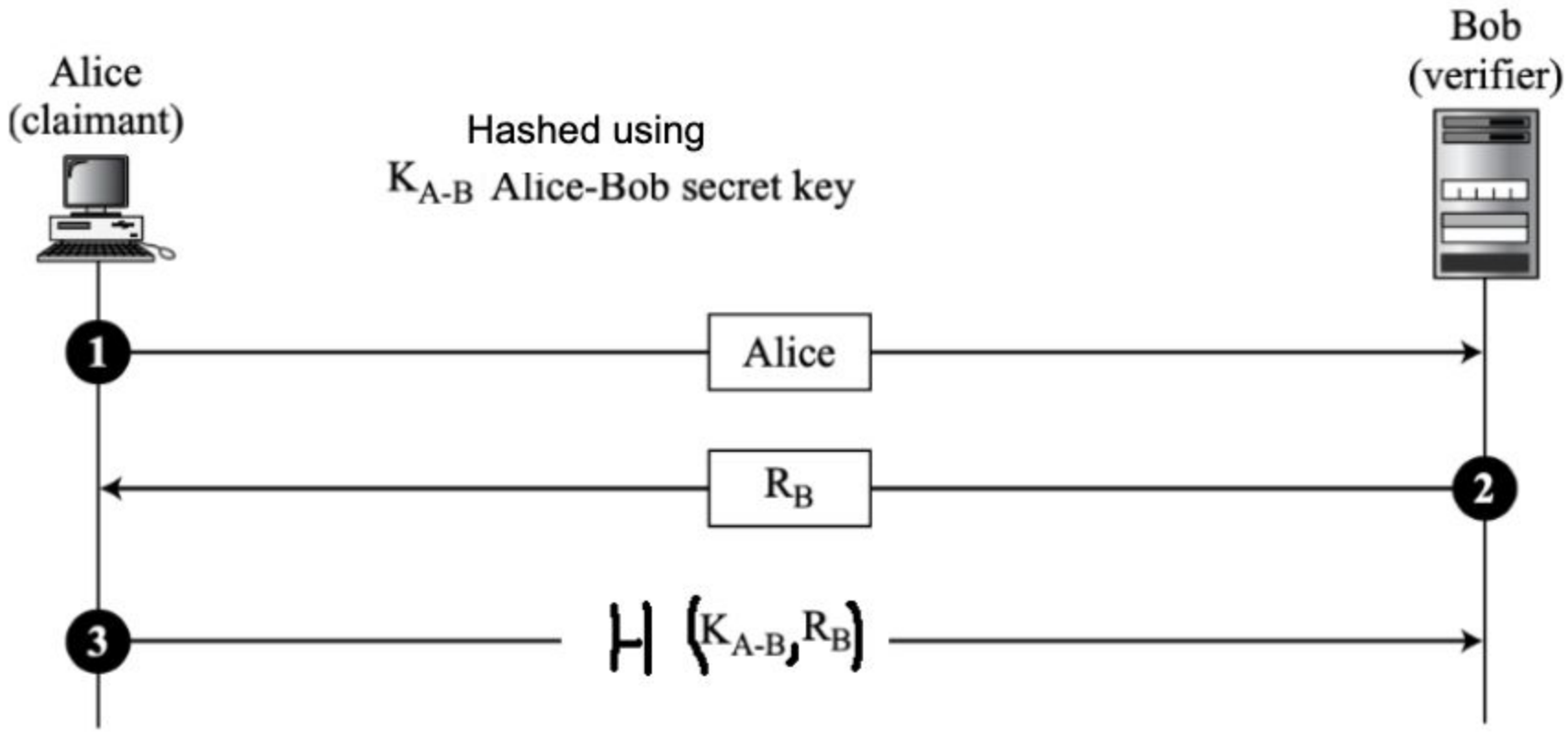
14.5 *Nonce challenge*



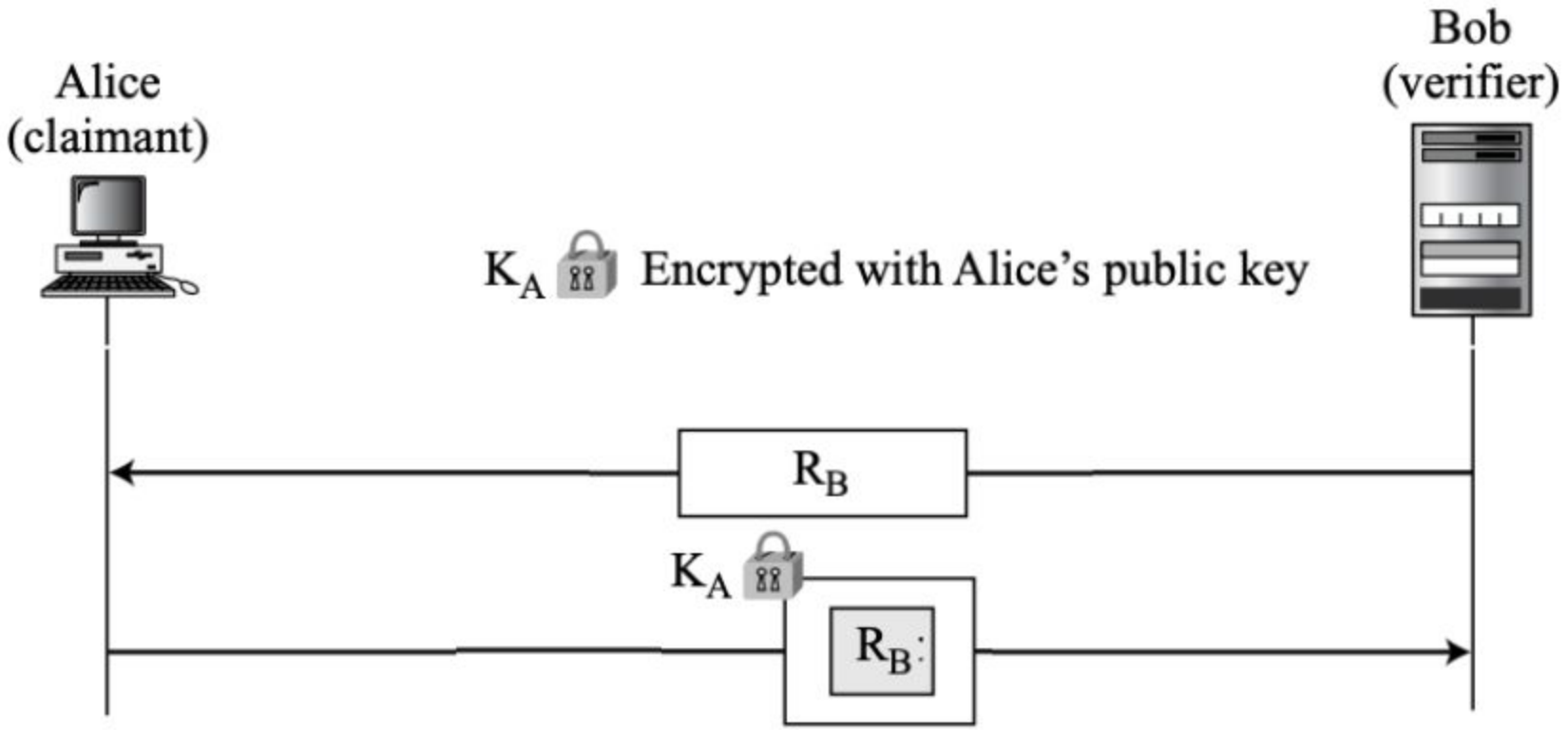
Symmetric Decryption



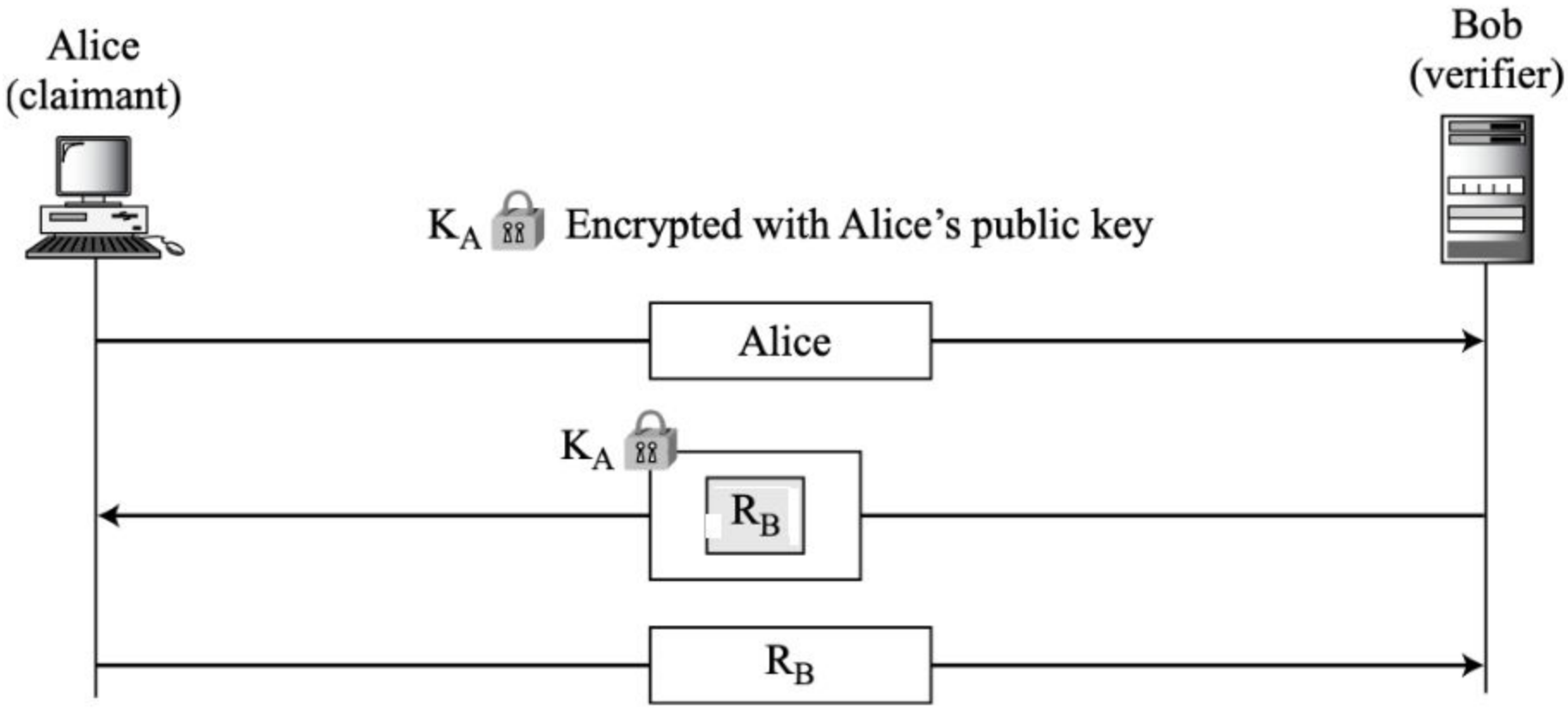
Symmetric Hash



Asymmetric Encryption

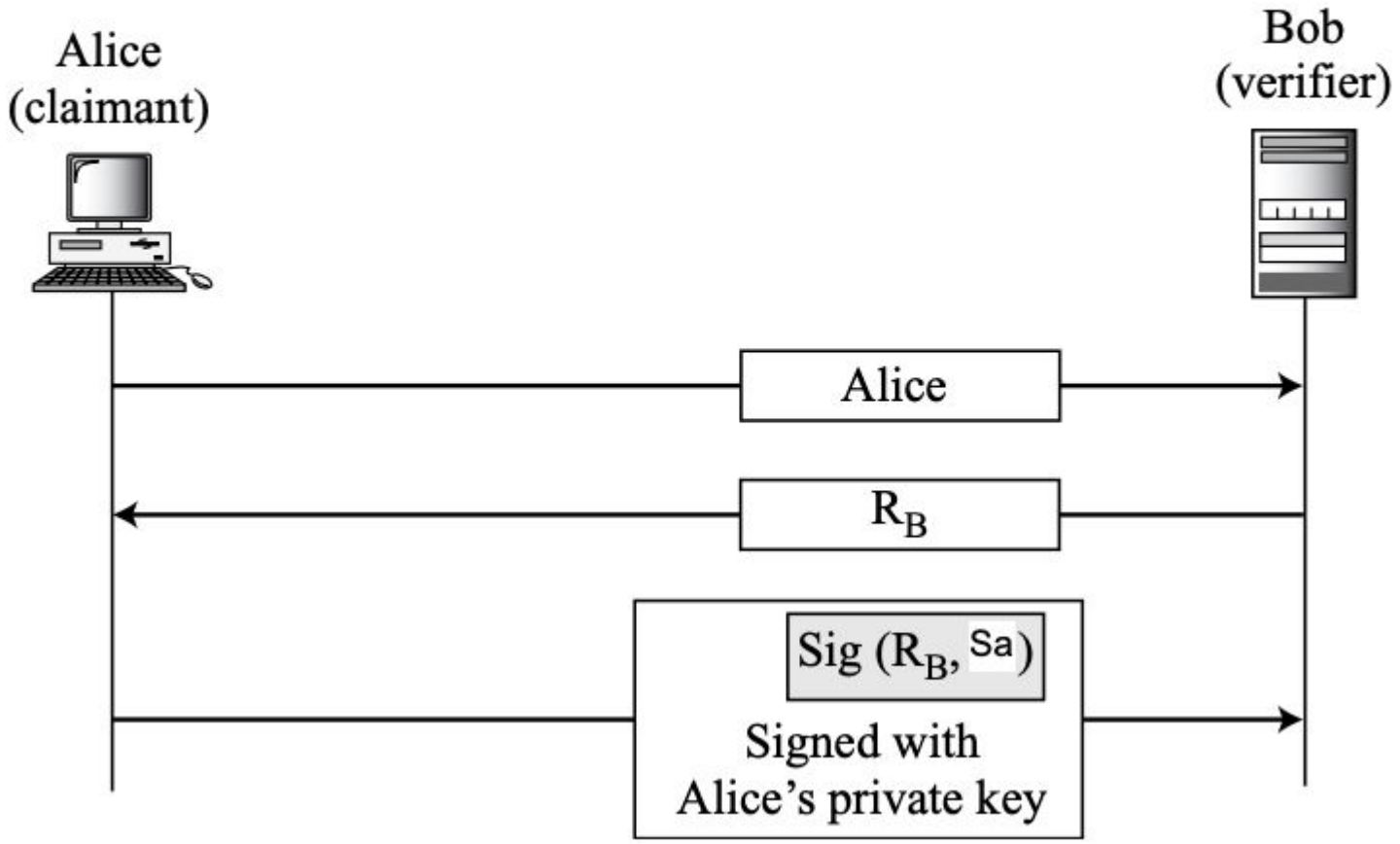


Asvemteric Decryption



Asymmetric Signature

Digital signature,



Challenge response

Shares secret with verifier

as pre shared key in symmetric algorithms

as ability to use the secret key for chosen text in asymmetric algorithms

e.g.

1. **For forging signature on message of choice**
2. **For cryptanalysis with chosen plaintext**

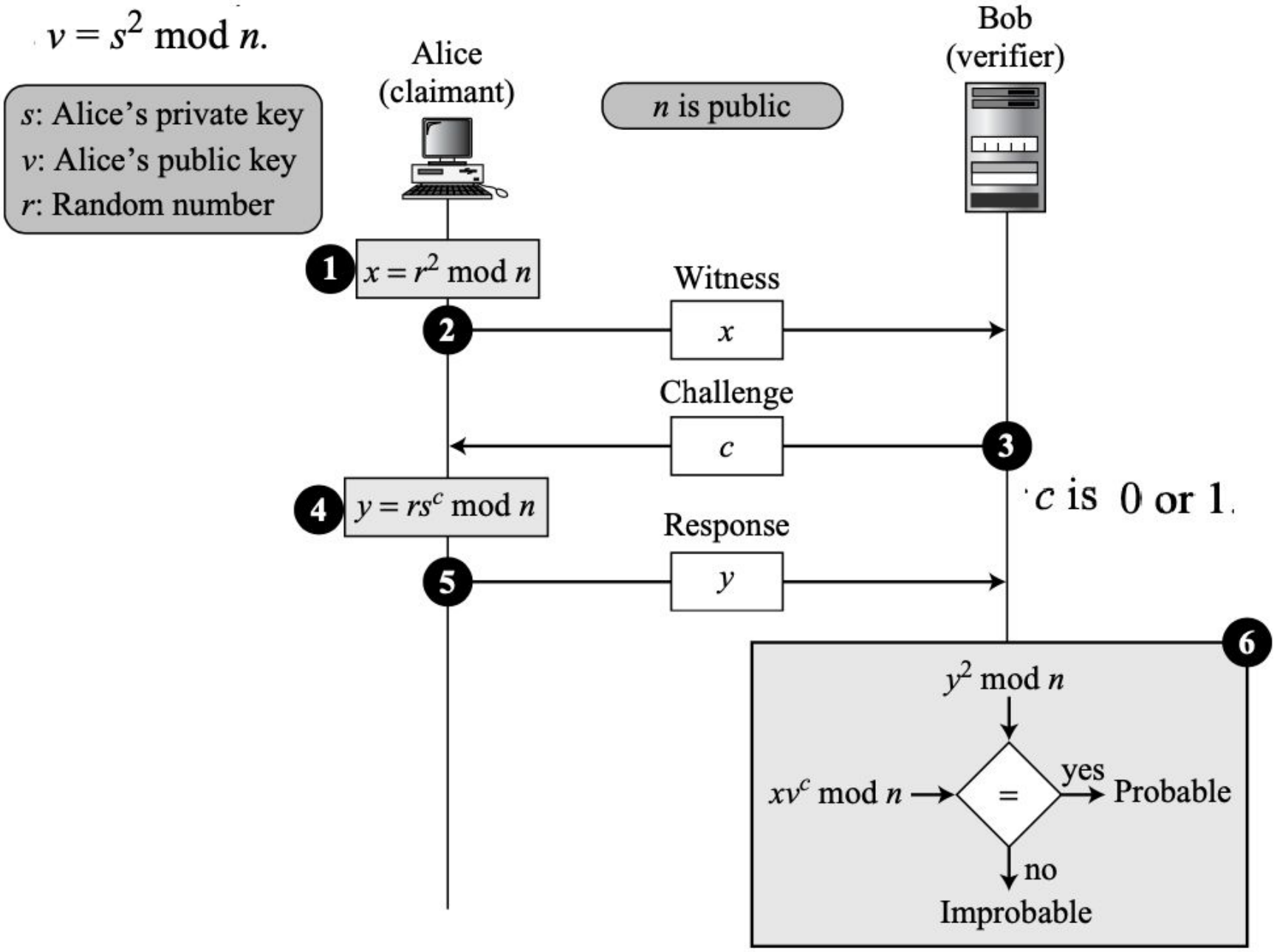
Zero Knowledge proof

Verify knowledge of a secret without revealing the secret with the verifier

or without giving verifier the ability to use the secret for a chosen text

all transcripts shared by claimant must be produceable by a simulator with sufficient probability **(without access to secret key)**

Fiat-Shamir protocol



Analysis

Verifier does not chose text ® to use the secret (s).
it doesn't even know r beforehand

Any simulator can pass with 50 % chance by selecting
r, x=r^2 mod N, and y =r mod N (passing for c=0)
or
y, x= (Y^2/v) mod N (passing for c=1)

For $c = 0$: $x = r^2 \bmod N, \quad y = r \bmod N$

For $c = 1$: Chose any y: $x = (y^2/v) \bmod N$

the attacker fails after repeated rounds due to decreasing probability of winning multiple rounds due to random c {0,1)